Adapter board fixes for the SAR ADC measurement

`IHP130_ASICONE_ADP_QFN48_0p5Q_Adapter`

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1 Problem

The characterization of the ADC (report `software/mso/results/sweep_20261022`) found three problems on the adapter board, none of them in the chip:

1. **Front-end bandwidth.** The single-ended to differential converter (TI SBOA216 [1]) is built with an LMV358 (GBW about 1 MHz, slew rate about 1 V/µs [2]). v_p passes one op-amp at unity gain, v_n that op-amp and an inverting stage with a noise gain of 2, so the two inputs of the ADC have different bandwidths. At 650 kHz the amplitude at the ADC falls to 0.50 of its low-frequency value, the input common mode moves by about half of the differential signal at the signal frequency, and the op-amps are close to their slew limit and distort (H2 rises from −45 dBc to −24 dBc; the transistor-level simulation shows that the ADC itself rejects the common-mode ripple, so this H2 is the op-amps). The ENOB falls from 3.8 to 3.1 bits.

2. **No decoupling on the references.** VREFP (pin 45) and VREFN (pin 44) come from the other LMV358 through jumpers JP6 and JP7, with no capacitor at the chip. The capacitive DAC draws charge from them at every bit decision (every 100 ns), much faster than a 1 MHz op-amp can correct, so the references may move with the code. This can produce the bit-weight errors seen in the static test (up to 8 %); the measurement cannot tell them apart from errors of the chip itself.

3. **Unfiltered 0.6 V.** The reference of the inverting stage is the divider R1/R2 on the 1.2 V LDO, without a capacitor. Its noise appears doubled on v_n only.

2 Rework of the present board

These changes use the existing footprints (SOIC-8 for U4 and U5, 1206 passives) and need no new board.

2.1 Replace U4 and U5 with OPA2365

The OPA2365 [3] is a dual op-amp with the standard pinout of the LMV358 in SOIC-8: 50 MHz GBW, 25 V/µs, rail-to-rail input and output, 2.2 V to 5.5 V supply (the board gives 3.3 V), 4.5 nV/$\sqrt{\text{Hz}}$. The signals of the front end (0.4 V to 0.8 V) and the references are well inside its input and output ranges. With it, both paths of the front end have bandwidths of tens of MHz, the difference between them becomes negligible at the frequencies of interest, and the slew rate needed (0.73 V/µs at 650 kHz) is 30 times below the limit. Alternative: OPA2350 (38 MHz, 22 V/µs, also SOIC-8 and rail-to-rail).

Both are unity-gain stable but drive capacitive loads: the front end drives the RC network through 47.5 Ω (as now) and the reference buffers drive 1 µF through 10 Ω (next section). These are the usual isolation resistors for capacitive loads; check the phase margin with the TI SPICE model of the part before the rework.

2.2 Decouple the references

- Remove the shunts of JP6 and JP7 and solder a 10 Ω resistor across each header instead.
- On the chip side of each jumper (pin B, net `adc_vrefp` or `adc_vrefn`), solder 1 µF and 100 nF X7R capacitors to the nearest ground, as close to pins 45 and 44 as the board allows.

The capacitors supply the charge of each bit decision; the 10 Ω keeps the op-amp stable and, with 1.1 µF, recharges them with a time constant of 11 µs. The average current of the DAC is about 1 µA, so the drop on the resistor is 10 µV, far below an LSB (25 mV differential). The feedback stays at the op-amp output, so the potentiometers keep setting the same voltages.

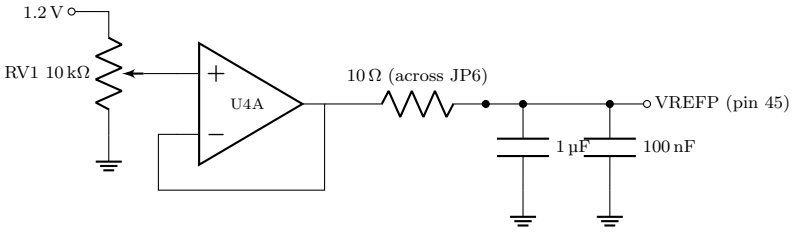


Figure 1: Reference buffer after the rework (U4A: OPA2365; VREFN is the same with RV2, U4B and JP7).

2.3 Filter the 0.6 V reference of the inverting stage

Solder 100 nF across R2 (from the non-inverting input of U5B to ground). With R1‖R2 = 5 kΩ the corner is at 320 Hz, which removes the noise of the LDO from v_n and keeps it from the comparator decisions.

2.4 Capacitors of the RC network

C19, C20 and C21 carry the signal: they must be C0G (NP0). The dielectric of the parts on the BOM (LCSC C541591 and C2909429) could not be confirmed; if they are X7R, replace them, because the capacitance of X7R depends on the voltage and distorts the signal. Optionally, C19 can go from 1.2 nF to 470 pF C0G: the differential pole moves from 1.33 MHz to 3.2 MHz (2 % of loss at 650 kHz instead of 10 %) while the reservoir per side (1.06 nF) is still 1000 times the sampling capacitance of the ADC (the kick of a sample is below 0.4 mV) and recovers with 50 ns, half of the sampling time.

2.5 Optional: matched R3 and R4

The gain of the inverting stage is R4/R3; a mismatch makes v_n larger or smaller than v_p and leaves a small common-mode ripple at all frequencies and adds to the gain and offset error of the measurement. The ADC rejects such a ripple well, so this is a refinement, not a fix. Two 0.1 % resistors of the same value, for example 2 kΩ, reduce it tenfold.

2.6 Expected result

Figure 2 and Table 1 apply the linear model of the measurement report (which reproduces the measured roll-off with the LMV358) to the proposed parts.

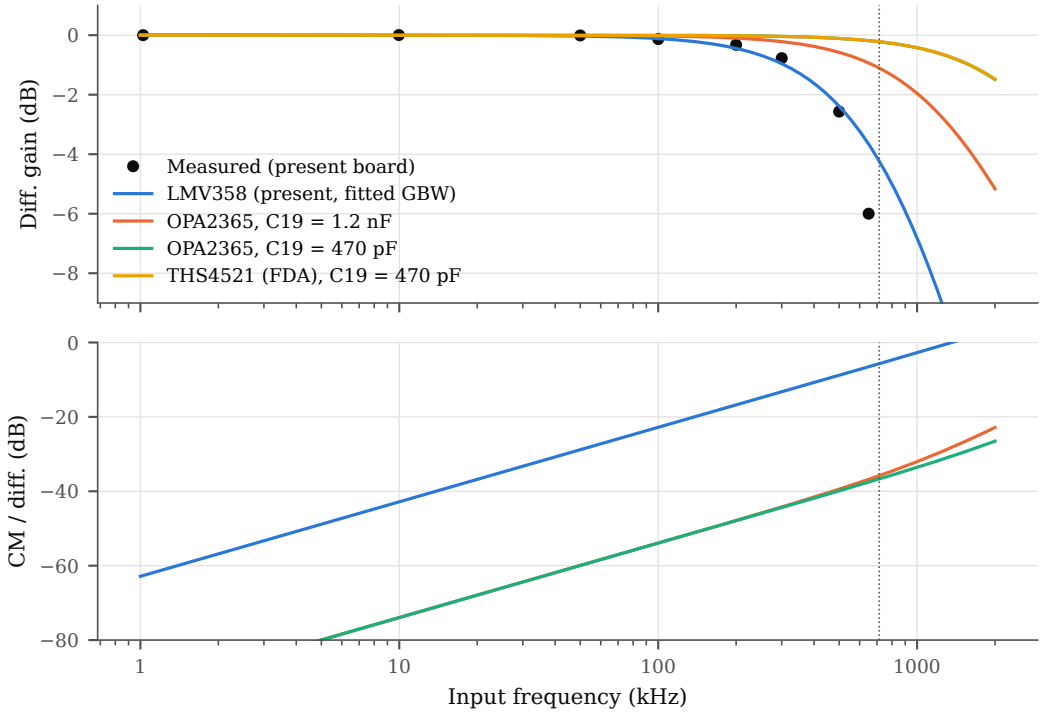


Figure 2: Differential gain and common-mode ripple at the ADC pins with the present and the proposed front ends (model; dotted line: Nyquist frequency, 714 kHz).

Front end	Gain, 300 kHz	Gain, 650 kHz	CM/diff., 650 kHz (%)
LMV358 (present, fitted GBW)	0.90	0.66	47.1
OPA2365, C19 = 1.2 nF	0.98	0.90	1.4
OPA2365, C19 = 470 pF	1.00	0.98	1.3
THS4521 (FDA), C19 = 470 pF	1.00	0.98	0.0

Table 1: Gain relative to low frequency and common-mode ripple (model). The fully differential amplifier has no ripple from the paths; what remains depends on its resistor matching.

With the OPA2365 the amplitude at 650 kHz goes from 0.66 to 0.90 (0.98 with C19 = 470 pF) and the common-mode ripple from 47.1 % to 1.4 %: the front end stops limiting the sweep, and the ENOB at high frequency should approach the low-frequency value, set by the static linearity. The reference decoupling can only be evaluated by measurement: if the bit-weight errors of the ramp test shrink after it, they came from the references; if not, they are in the chip.

3 Next revision of the board

For a new board, a fully differential amplifier is the better front end: one stage drives both outputs through the same path, so they stay opposite at any frequency, and its output common mode is set directly by the VOCM pin. The THS4521 [4] (95 MHz GBW, 490 V/µs, rail-to-rail output, input range including ground, 2.5 V to 5.5 V supply, SOIC-8 or VSSOP-8) fits the 3.3 V supply. For the same transfer as now, $v_d = 2(v_{in} - 0.6\,\mathrm{V})$, use $R_f = 2R_g$ (for example 1 kΩ and 499 Ω), drive the gain resistor of the non-inverting input from v_{in} and the other from the decoupled 0.6 V, and connect VOCM to the same 0.6 V. Keep the 47.5 Ω resistors and the C0G network at the ADC.

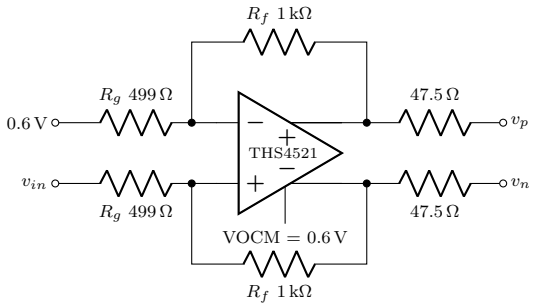


Figure 3: Fully differential front end for a new board (the C0G network at the ADC as before). $v_p - v_n = (R_f/R_g)(v_{in} - 0.6\,\mathrm{V})$: v_{in} goes to the non-inverting side so that the positive output is in phase with it.

4 Parts

Ref.	Change	Part	LCSC
U4, U5	LMV358 → OPA2365	TI OPA2365AIDR, SOIC-8	C130011
	(alternative)	TI OPA2350UA/2K5, SOIC-8	C55904
JP6, JP7	shunt → 10 Ω	any 1206 1 % resistor	–
new	1 µF at pins 44, 45	Samsung CL31B105KAHNNNE, 1206 X7R 25 V	C28322
new	100 nF at pins 44, 45 and across R2	as C1 on the BOM, 1206	C697702
C19–C21	C0G if not already	1206 C0G, 50 V, 5 %	–
R3, R4	optional, matched	RESI PTFR1206B2K00N9, 2 kΩ 0.1 %	C2692717
new board	fully differential amplifier	TI THS4521IDR, SOIC-8	C16092

LCSC numbers from the LCSC listings in October 2026; check stock before ordering, and confirm C28322 against the Samsung part number. LCSC marks the THS4521IDR as not recommended for new designs while TI lists it as active.

5 Verification

After the rework, repeat `adc_sweep.sh` with the same settings and rebuild the report:

- the ADC amplitude should stay within a few percent of the 1 kHz value up to 650 kHz, and H2 should stay near its low-frequency level;
- the ENOB at high frequency should approach the low-frequency value;
- compare the DNL/INL of the ramp before and after the reference decoupling.

References

[1] Texas Instruments, *Single-Ended Input to Differential Output Circuit*, Analog Engineer's Circuit SBOA216B, rev. Oct. 2024. `https://www.ti.com/lit/pdf/sboa216`

[2] Texas Instruments, *LMV321, LMV358, LMV324 Low-Voltage Rail-to-Rail Output Operational Amplifiers*, datasheet SLOS263Y, rev. Aug. 2023. `https://www.ti.com/lit/gpn/lmv358`

[3] Texas Instruments, *OPA365, OPA2365 50-MHz, Zerø-Crossover, Low-Distortion, High CMRR, RRI/O, Single-Supply Operational Amplifiers*, datasheet SBOS365G, rev. May 2023. `https://www.ti.com/product/OPA2365`

[4] Texas Instruments, *THS452x Very Low Power, Negative Rail Input, Rail-to-Rail Output, Fully Differential Amplifier*, datasheet SBOS458H, rev. June 2015. `https://www.ti.com/product/THS4521`